

CSE 2025

Remote Sensing C TEST

Written by Aaron Kou, Ann Arbor Pioneer High School

This test consists of 50 multiple-choice questions worth two points each, and 6 multi-part free-response questions worth varying amounts of points. There are a total of 196 points.

Multiple-choice; one correct response; 2 points each

1. The greenhouse effect is caused by the interaction between gas molecules and which type of electromagnetic radiation?
    a. Radio
    b. Microwave
    c. Infrared
    d. Ultraviolet

2. Which of the following correctly arranges the albedos of the surfaces, from least to greatest?
    a. Open ocean, deciduous forest, snow
    b. Deciduous forest, open ocean, snow
    c. Deciduous forest, snow, open ocean
    d. Snow, open ocean, deciduous forest

3. What is the trend in Earth's albedo and the largest cause of this trend?
    a. Decreasing; melting of ice and snow
    b. Decreasing; land use changes
    c. Increasing; melting of ice and snow
    d. Increasing; land use changes

4. Which of the following combinations of cloud conditions will create the smallest daily temperature range?
    a. Clear skies in the day and clear skies at night
    b. Clear skies in the day and cloudy skies at night
    c. Cloudy skies in the day and clear skies at night
    d. Cloudy skies in the day and cloudy skies at night

5. For a perfect blackbody, the radiant exitance is _____ proportional to the _____ power of its temperature.
    a. Directly; third
    b. Directly; fourth
    c. Inversely; third
    d. Inversely; fourth

6. Which of the following frequencies would be expected of light coming from a LiDAR sensor?
    a. 3×10^5 Hz
    b. 3×10^8 Hz
    c. 3×10^{11} Hz
    d. 3×10^{14} Hz

7. Which device would be best used to measure the distance to the ground on an airplane?
    a. Accelerometer
    b. Crookes radiometer
    c. Hyperspectral radiometer
    d. Radar altimeter

8. The surface temperature of the Sun is around 5700 K. What range of the electromagnetic spectrum does its peak wavelength lie in?
    a. Visible
    b. Ultraviolet
    c. X-ray
    d. Gamma ray

9. Which of the following is <u>not</u> a common application of infrared imagers?
    a. Detection of gravitational anomalies
    b. Land surface temperature monitoring
    c. Vegetation management
    d. Wildfire detection

10. Which combination of aerosols will create the greatest warming in the troposphere?
    a. Black carbon and ozone
    b. Black carbon and sulfur dioxide
    c. Dust and ozone
    d. Dust and sulfur dioxide

11. Which of the following equations yields a satellite's height above the ground, *c* being the speed of light and *t* being the time between successive pulses?
    a. $2ct$
    b. $ct/2$
    c. $2c/t$
    d. $c/2t$

12. A photon with energy 2.5 eV interacts with a particle whose diameter is 0.5 μm. What type of scattering will occur?
    a. Compton
    b. Geometric
    c. Mie
    d. Rayleigh

13. Which of the following is <u>not</u> a carbon trap?
    a. Ocean uptake
    b. Photosynthesis
    c. Transpiration
    d. Weathering of carbonate rocks

14. Which of the following is considered a trace gas in the atmosphere?
    a. Ar
    b. CO_2
    c. O_2
    d. N_2

15. Which of the following best describes passive sensors?
    a. Are often less expensive
    b. Can operate at night-time
    c. Can penetrate dense cloud cover
    d. Require continuous power consumption

16. Increased concentration of aerosols will create a _____ number of _____ water droplets.
    a. Fewer; larger
    b. Fewer; smaller
    c. Greater; larger
    d. Greater; smaller

17. Which of the following cloud types typically has the greatest warming effect?
    a. Cirrus
    b. Cumulonimbus
    c. Cumulus
    d. Stratus

18. According to Planck's radiation law, which of the following will decrease when the temperature of a blackbody increases?
    a. Albedo
    b. Peak wavelength of the spectrum
    c. Spectral energy density
    d. Total radiated energy

19. Which of the following correctly ranks the GWP-100 of the four gasses, from least to greatest?
    a. CO_2, CH_4, N_2O, SF_6
    b. CO_2, N_2O, CH_4, SF_6
    c. SF_6, N_2O, CH_4, CO_2
    d. SF_6, CH_4, N_2O, CO_2

20. Which instrument aboard GOES-16 is used to measure air temperature?
    a. ABI
    b. GLM
    c. MAG
    d. SEISS

21. Which of the following is <u>not</u> a limitation of remote sensing data compared to climate models?
    a. Atmospheric conditions
    b. Data processing
    c. Sensor resolutions
    d. Simplifying assumptions

22. Approximately what percent of carbon dioxide produced by humans is absorbed by the ocean?
    a. 10%
    b. 20%
    c. 30%
    d. 40%

23. Which of the following best describes the difference between spectrometers and radiometers?
    a. Spectrometers measure visible light, while radiometers measure radio waves
    b. Spectrometers are passive sensors, while radiometers are active sensors
    c. Spectrometers rely on specular reflection, while radiometers rely on diffuse reflection
    d. Spectrometers can distinguish multiple wavelengths, while radiometers cannot

24. When modeling sea ice, researchers observe that the rate of melting in the model is higher than what is observed in real life. Which of the following parameters of the model may have been overemphasized?
    a. Albedo of ice
    b. Evaporation
    c. Incoming solar radiation
    d. Positive sensible heat flux

25. Which of the following resources is most easily detected using gravitational methods?
    a. Groundwater
    b. Iron
    c. Petroleum
    d. Vegetation

26. ATMS and CrIS are both aboard _____ and CrIS operates at _____ wavelengths.
    a. CALIPSO; shorter
    b. CALIPSO; longer
    c. Suomi NPP; shorter
    d. Suomi NPP; longer

27. The refractive index of ice is 1.31. What is the speed of light through ice?
    a. 1.75×10^8 m/s
    b. 2.29×10^8 m/s
    c. 3.00×10^8 m/s
    d. 3.93×10^8 m/s

28. The device in Image 1 is put in a freezer. Which side of the vane cools more rapidly, and (looking from above) what direction does the device turn?
    a. Black; clockwise
    b. Black; counter-clockwise
    c. White; clockwise
    d. White; counter-clockwise

29. Image 2 depicts data from a radar. Which of the following would not be expected in the area?
    a. Advection fog
    b. Derecho
    c. Squall line
    d. Straight-line winds

30. According to Image 3, at which wavelength is the attenuation coefficient for Earth's atmosphere highest?
    a. 4 microns
    b. 6 microns
    c. 8 microns
    d. 10 microns

31. Using Image 3, which of the following movements of the water molecule causes the low transmittance in the spectrum?
    a. Photodissociation
    b. Photoionization
    c. Rotation
    d. Vibration

32. Which of the following is not a conclusion that can be drawn from Image 4?
    a. Some areas have cooled in the past 30 years, despite warming before 1994
    b. The Northern Hemisphere is warming faster than the Southern Hemisphere
    c. Global warming has accelerated in the past 30 years
    d. The continents are warming, while the oceans are cooling

33. What is the most likely reason the Arctic is dark red in the lower map in Image 4?
    a. More frequent thunderstorms
    b. Deposition of black soot
    c. Reduced albedo
    d. Increased sulfur dioxide emissions

34. Image 5 depicts net flux, with purple indicating a negative flux and orange indicating a positive flux. Which instrument/satellite collected this data?
    a. AURA
    b. CERES
    c. GRACE
    d. TOPEX-Poseidon

35. In which month was Image 5 taken?
    a. January
    b. April
    c. July
    d. October

36. Image 6 depicts cloud cover data from which two satellites?
    a. Aqua and Aura
    b. Aqua and Terra
    c. CloudSat and Aura
    d. CloudSat and Terra

37. In Image 6, does the dark blue correspond to a higher or lower cloud cover?
    a. Higher
    b. Lower

38. Which of the following processes likely causes the white band across the equatorial Pacific in Image 6?
    a. Convergence and uplift of surface air
    b. Convergence and descent of air aloft
    c. Divergence and uplift of surface air
    d. Divergence and descent of air aloft

39. Using Image 7, approximately what value does the green color correspond to?
    a. 0.004%
    b. 0.04%
    c. 0.4%
    d. 4%

40. In Image 7, the red spot over China is a sign of _____, and the spots over South America and southern Africa are a sign of _____.
    a. Agriculture; agriculture
    b. Agriculture; industry
    c. Industry; agriculture
    d. Industry; industry

41. Image 7 was taken from October to November What trend would be expected as time passes towards December and January?
    a. Increased CO_2 in the Northern Hemisphere; increased CO_2 in the Southern Hemisphere
    b. Increased CO_2 in the Northern Hemisphere; decreased CO_2 in the Southern Hemisphere
    c. Decreased CO_2 in the Northern Hemisphere; increased CO_2 in the Southern Hemisphere
    d. Decreased CO_2 in the Northern Hemisphere; decreased CO_2 in the Southern Hemisphere

42. Image 8 comes from data taken in southern Texas. Outlying measurements indicate what?
    a. Exceptionally clear skies
    b. Extensive cloud cover
    c. High temperatures
    d. Smoke, dust, and smog

43. True or False: Aerosol optical depth can be higher than 1.
    a. True
    b. False

44. What is the most likely cause of the annual peaks during the summer in Image 8?
    a. California wildfires
    b. Gulf of Mexico eutrophication events
    c. Saharan dust storms
    d. Supercell thunderstorms

45. What compound/s is/are being observed in Image 9, and which satellite took the data?
    a. CFCs; Aura
    b. CFCs; GOES
    c. Ozone; Aura
    d. Ozone; GOES

46. In Image 9, the darkest red area has a value of around 100. What are the units for this image, and does the red color indicate a higher or lower value?
    a. Dobson units; higher
    b. Dobson units; lower
    c. ppm; higher
    d. ppm; lower

47. Which piece of legislation led to the trend observed between Images 9b and 9c?
    a. Clean Air Act
    b. Kyoto Protocol
    c. Montreal Protocol
    d. Paris Agreement

48. What value is being measured in Image 10?
    a. Global mean CO_2 concentration
    b. Global mean land surface temperature
    c. Global mean sea surface temperature
    d. Global mean sea level

49. What are the units of the y-axis in Image 10?
    a. cm
    b. °F
    c. mm
    d. ppm

50. What are the most direct causes of the trend shown in Image 10?
    a. Thermal expansion of water and the melting of land ice and glaciers
    b. Thermal expansion of water and the melting of sea ice
    c. Combustion of fossil fuels and agriculture
    d. Combustion of fossils fuels and deforestation

Free-response; point values are indicated in parentheses

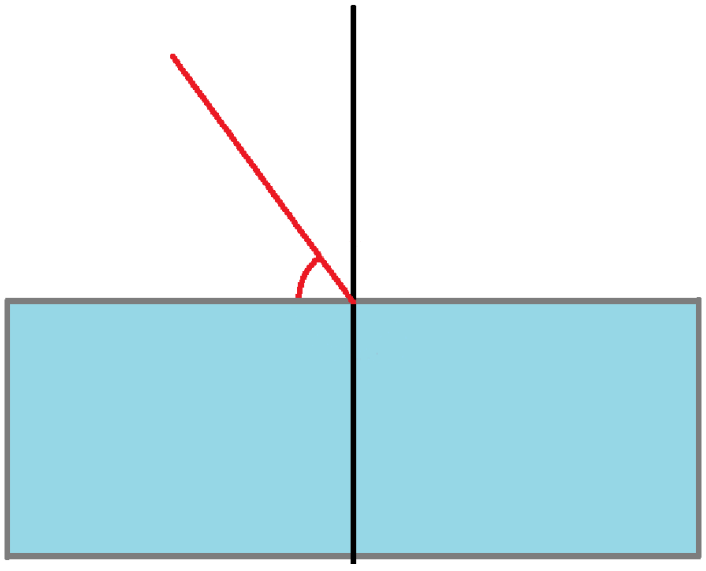
51. A ray of red light travelling through air (n = 1.00) strikes a surface of water (n = 1.33) at an angle of 53° from the horizontal. Some of the light reflects specularly off of the surface, while some of the light passes into the water and refracts.
    a. (2) With a solid line, draw, on the diagram at the bottom of the page, the path of the light that reflects specularly off of the surface and indicate the angle from the horizontal.
    b. (5) For the light that passes into the water, calculate the angle from the horizontal that the ray makes. Show your work.

    c. (2) Using a dashed line, draw the ray from part (b) on the diagram.
    d. (2) Determine the smaller of the two angles formed by the rays drawn in parts (a) and (c).

    e. (3) How would the angle calculated in part (d) change if the water was replaced by a medium with a higher index of refraction?

    f. (5) Through water, red light has an extinction coefficient of 0.6 m^{-1}. Using the Beer-Lambert Law, determine the distance that red light travels before reaching 1% of its original intensity.

    g. (3) Using the angle calculated in part (b) and the distance calculated in part (f), determine the depth that the light attains before reaching 1% of its original intensity.



52. Image 11 depicts a color-inverted image from NASA, showing the distribution of aerosols across the globe on August 23, 2018. Notice the three distinct colors: light blue, dark green, and brown.
    a. (2) List two of the four satellites that data from this image may have come from.

    b. (5) One of the colors corresponds to sea spray. Identify the color, and use a piece of evidence from the image to support your choice.

    c. (5) One of the colors corresponds to black carbon particles. Identify the color, and use a piece of evidence from the image to support your choice.

    d. (5) One of the colors corresponds to dust. Identify the color, and use a piece of evidence from the image to support your choice.

53. Image 12 depicts NO_2 levels over Africa during the dry season, with the blue color indicating a decrease in NO_2 levels.
    a. (2) What is the main source of nitrogen dioxide in the atmosphere?

    b. (3) Does the trend in the image represent an improvement or worsening of air quality? Briefly explain.

    c. (3) What change in land use likely led to this trend?

    d. (4) The trend shown in the image was unexpected by most scientists. Considering your answer to part (a), explain why.

54. The ocean is the world's largest carbon sink.
    a. (3) Describe the biological process by which carbon is trapped in the ocean.

    b. (3) Describe the physical process by which carbon is trapped in the ocean.

    c. (3) Is CO_2 more likely to be trapped at higher or lower latitudes? Explain.

    d. (5) Using your answer to part (c), describe the relationship between a warming atmosphere and the effectiveness of the ocean as a carbon sink. Is this a positive or negative feedback loop?

55. This question pertains to the twin GRACE satellites, sometimes called Tom and Jerry!
    a. (2) What does GRACE stand for?

    b. (1) What do the two satellites detect?

    c. (6) Describe the physics of what happens before, during and after Jerry (the leading satellite) and Tom (the following satellite) pass over a positive [thing you identified in part (b)].

56. This question explores the theory behind planetary energy balance for a simplified planet with and without a simple atmosphere. Let R be the amount of shortwave radiation coming toward the planet, a be the albedo of the planet, T be the temperature of the planet, and σ be the Stefan-Boltzmann constant, equal to 5.67×10^{-8} $W/(m^2 \cdot K^4)$.

    a. (3) In terms of R, a, T, and σ (some values may not be used), determine the amount of radiation absorbed by the surface of the planet.

    b. (3) In terms of R, a, T, and σ (some values may not be used), determine the amount of radiation reflected and emitted by the planet.

    c. (4) Derive an expression for the equilibrium temperature of the planet in terms of R, a, and σ.

Now, let's introduce an atmosphere that is transparent to shortwave radiation but absorbs and re-emits longwave radiation. Let T_A be the temperature of the atmosphere, and T_S be the temperature at the surface.

    d. (4) Write an expression equating the incoming and outgoing radiation at the top of the atmosphere.

    e. (4) Repeat part (d), but this time at the ground.

    f. (4) Is T_s greater than, less than, or equal to T_A? Show why, using your answers to parts (d) and (e).